WORKED SOLUTION

STEP III 2001, Question 1

Pure • Maclaurin Series

STEP III 2001, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q1

The key idea

The function $y = \ln(x + \sqrt{x^2+1})$ is $\sinh^{-1} x$, and its first derivative is far simpler than the function itself. Squaring that derivative gives a relation with no square roots, $(x^2+1)(y')^2 = 1$, and differentiating it gives the $n = 0$ case of the recurrence. Each further differentiation moves the recurrence up one step, which is exactly what an induction needs. At $x = 0$ the x term disappears and the recurrence links $y^{(n+2)}(0)$ straight to $y^{(n)}(0)$, so all the Maclaurin coefficients come almost for free.

Solution

The derivative

Write $s = \sqrt{x^2+1}$, so that $\dfrac{\mathrm{d}s}{\mathrm{d}x} = \dfrac{x}{s}$. Note that $x + s > 0$ for every real x, since $s > \sqrt{x^2} = |x|$, so the logarithm is defined everywhere. By the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x+s}\left(1 + \frac{x}{s}\right) = \frac{1}{x+s} \cdot \frac{s+x}{s} = \frac{1}{\sqrt{x^2+1}},$$

as required.

The induction

Let $P(n)$ be the statement

$$(x^2+1)\,y^{(n+2)} + (2n+1)\,x\,y^{(n+1)} + n^2\,y^{(n)} = 0 \quad \text{for all real } x.$$

Base case, $n = 0$. We need $(x^2+1)y'' + xy' = 0$. From the first part, $(x^2+1)(y')^2 = 1$. Differentiating with respect to x,

$$2x(y')^2 + (x^2+1) \cdot 2y'y'' = 0.$$

Since $y' = 1/\sqrt{x^2+1}$ is never zero, we may divide by $2y'$, which gives $(x^2+1)y'' + xy' = 0$. That is $P(0)$. (Alternatively, differentiate $y' = (x^2+1)^{-1/2}$ directly: $y'' = -x(x^2+1)^{-3/2}$, so $(x^2+1)y'' = -x(x^2+1)^{-1/2} = -xy'$.)

Inductive step. Suppose $P(k)$ holds for some $k \geq 0$:

$$(x^2+1)\,y^{(k+2)} + (2k+1)\,x\,y^{(k+1)} + k^2\,y^{(k)} = 0.$$

This holds for all x, so we may differentiate it. Using the product rule on the first two terms,

$$2x\,y^{(k+2)} + (x^2+1)\,y^{(k+3)} + (2k+1)\,y^{(k+1)} + (2k+1)\,x\,y^{(k+2)} + k^2\,y^{(k+1)} = 0.$$

Collecting terms:

$$(x^2+1)\,y^{(k+3)} + (2k+3)\,x\,y^{(k+2)} + (k^2+2k+1)\,y^{(k+1)} = 0,$$

and since $2k+3 = 2(k+1)+1$ and $k^2+2k+1 = (k+1)^2$, this is precisely $P(k+1)$.

Ask yourself Why does the coefficient of $y^{(k+2)}$ go up by exactly 2?

Differentiating (x^2+1) in the first term throws off a $2x\,y^{(k+2)}$, which joins the $(2k+1)x\,y^{(k+2)}$ from the second term. Similarly the $(2k+1)$ from differentiating the x in the second term joins k^2, and k^2+2k+1 is a perfect square. The recurrence is built to reproduce itself.

Since $P(0)$ holds and $P(k) \Rightarrow P(k+1)$ for every $k \geq 0$, by induction $P(n)$ holds for all $n \geq 0$.

The Maclaurin series

Put $x = 0$ in $P(n)$. The middle term vanishes, leaving

$$y^{(n+2)}(0) = -n^2\,y^{(n)}(0) \qquad (n \geq 0).$$

We need two starting values: $y(0) = \ln(0+1) = 0$ and $y'(0) = 1/\sqrt{1} = 1$.

Even derivatives. From $y(0) = 0$ the recurrence gives $y''(0) = 0$, then $y^{(4)}(0) = 0$, and so on: every even-order derivative vanishes at 0. (This also follows from y being an odd function: $y(-x) = \ln(\sqrt{x^2+1} - x) = -\ln(\sqrt{x^2+1} + x)$, because $(\sqrt{x^2+1} - x)(\sqrt{x^2+1} + x) = 1$.)

Odd derivatives. Starting from $y'(0) = 1$,

$$\begin{aligned} y'''(0) &= -1^2 \cdot y'(0) = -1, \\ y^{(5)}(0) &= -3^2 \cdot y'''(0) = 9, \\ y^{(7)}(0) &= -5^2 \cdot y^{(5)}(0) = -225. \end{aligned}$$

Ask yourself Which value of n gives $y^{(5)}(0)$ from $y'''(0)$?

To get $y^{(n+2)}$ with $n+2=5$ we need $n=3$, so the factor is $-3^2=-9$, not -5^2. The factor uses the order of the *lower* derivative.

The Maclaurin series $y=\sum y^{(k)}(0)\,x^k/k!$ is therefore

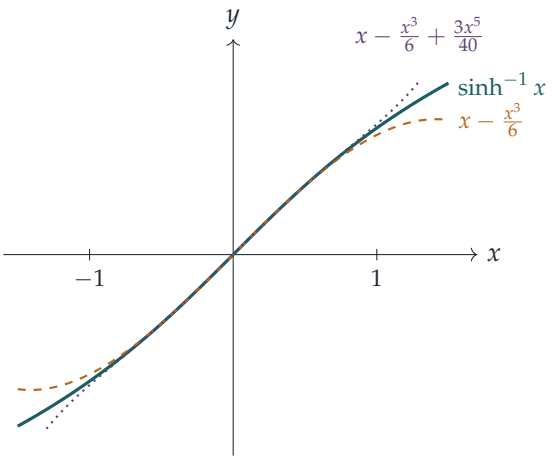
$$\begin{aligned} y &= x+\frac{-1}{3!}x^3+\frac{9}{5!}x^5+\frac{-225}{7!}x^7+\cdots \\ &= x-\frac{x^3}{6}+\frac{9}{120}x^5-\frac{225}{5040}x^7+\cdots \\ &= x-\frac{x^3}{6}+\frac{3x^5}{40}-\frac{5x^7}{112}+\cdots, \end{aligned}$$

which begins as required, and the next non-zero term is

$$-\frac{5x^7}{112}.$$

Another route. Integrating the binomial series $(1+x^2)^{-1/2}=1-\frac{1}{2}x^2+\frac{3}{8}x^4-\frac{5}{16}x^6+\cdots$ term by term (valid for $|x|<1$), with $y(0)=0$, gives $x-\frac{1}{6}x^3+\frac{3}{40}x^5-\frac{5}{112}x^7+\cdots$, which agrees.

The picture shows how the partial sums hug $y=\sinh^{-1}x$ near the origin, alternately overshooting and undershooting further out.



Checks

- At $x=0.1$, $\ln\left(0.1+\sqrt{1.01}\right)=0.0998340789$, while $x-\frac{x^3}{6}+\frac{3x^5}{40}-\frac{5x^7}{112}=0.0998340789$ as well; the difference is about 3×10^{-11}, the size of the next (x^9) term.
- The $n=0$ case is easy to test at $x=1$: $y'=1/\sqrt{2}$ and $y''=-1/(2\sqrt{2})$, so $2y''+y'=0$.

- The odd derivatives at 0 are $1, -1, 9, -225$, that is $(-1)^m\big(1 \cdot 3 \cdots (2m-1)\big)^2$: a pleasing pattern worth spotting.

Common slips

- Using the wrong n in $y^{(n+2)}(0) = -n^2 y^{(n)}(0)$, giving factors $-9, -25, -49$ instead of $-1, -9, -25$.
- Forgetting to divide by $k!$: the coefficient of x^7 is $-225/5040$, not -225.
- In the inductive step, missing the $2x\,y^{(k+2)}$ that comes from differentiating $x^2 + 1$.

Beyond STEP

A Taylor series is only useful where it converges, and this one converges exactly for $|x| \leq 1$, even though $\sinh^{-1} x$ is perfectly smooth on the whole real line. The reason lives in the complex plane: $1/\sqrt{1+z^2}$ blows up at $z = \pm\mathrm{i}$, at distance 1 from the origin, and that distance is the radius of convergence. Needham's *Visual Complex Analysis* explains this beautifully; for the real-variable story of Taylor's theorem and its remainder, see Abbott, *Understanding Analysis*.

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